

Text Books

1. J. J. Uicker, G. R. Pennock, and J. E. Shigley, Theory of Machines and Mechanism, Oxford Press (2009).
2. Dam B. Marghit, Mechanisms and Robotics analysis with MATLAB, Springer (2009).
3. T. Bevan, The Theory of Machines, (Pearson Edu. India) CBS Publishers (1944).

Reference Books

1. A. K. Ghosh and A. K. Mallik, Theory of Mechanisms and Machines, Affiliated East-West press Pvt. Ltd. (1993).
2. C. E. Wilson and J. P. Sandler, Kinematics and Dynamics of Machines, Pearson Publishers (2003).
3. B. Crossland and I. Morrison, Mechanics of Machines, Longman Publishers (1970).
4. J. Hannah and R. C. Stephens, Mechanics of Machines, Elementary Theory and Examples, International Ed (1970).
5. S. S. Rattan, Theory of Machines, Tata Mc Graw Hill (2012).

Evaluation Scheme

S.No	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	35
3.	Sessionals (May include assignments/Projects/Tutorials/Quizes/Lab Evaluations)	40

UME310 : Manufacturing Technology

L	T	P	Cr
2	0	2	3.0

Course Objective: The objective of this course is to introduce to the students different modes of solidification in metal casting and design of gating, riser system required for casting. The students will study the concept of utilization of heat during welding. The students will study bulk metal forming operations and calculate the force, power requirements during different forming and metal cutting operations. Further the students

will also study different manufacturing processes for plastics and composites.

Syllabus

Metal Cutting: Principles of orthogonal and oblique cutting, shear angle relationships, Forces in metal cutting, Tool life equation, Machinability, Milling, milling cutters and milling machines. Drilling, Grinding, grinding wheel selection, surface grinding.

Metal Forming: ; High energy rate forming processes, forging machines, forging design considerations, forging defects, Wire and bar drawing, extrusion. Sheet metal operations: Deep drawing, Perforation, Lancing , Nibbling.

Technology of Metal Casting: Design of Riser and gating system, investment casting, die casting, Shell moulding; electric furnace, Theory of solidification

Welding: Heat flow during welding, principles and applications of TIG and MIG welding, Submerged Arc Welding, Friction Stir Welding, Thermit Welding, welding defects.

Manufacturing of plastics & composites: Plastic Manufacturing: Materials, Additives, Extrusion; Injection Moulding, Extrusion Blow Moulding, Powder Metallurgy.

Laboratory Work: Experimental work pertaining to study & use of sand testing equipment, Experiments on casting process, performance on MIG welding, TIG welding exercises on horizontal & vertical milling machines, planer, shaper, centerless & surface grinders, profile cutting in vertical milling machine; milling indexing, various metal forming operation such as Nibbling , spinning etc.

Micro Project: Students in a group of 4/5 will carry out micro project on fabrication of a multi- operation job that includes machining, forming, casting and joining of dissimilar metals.

Course Learning Objectives (CLO)

The students will be able to:

1. analyse the various forces and energy required in metal cutting operations.
2. estimate the various forces required for forming operations.
3. design the gating and riser system for the casting process
4. evaluate the heat flow and select suitable welding technique for different applications
5. recognize the manufacturing processes for processing of plastics & composites.

Text Books

- 1 Rao, P.N., Manufacturing Technology: Foundry, Forming & Welding, Tata Mc-Graw Hill, New Delhi (2003)
- 2 Rao, P.N., Manufacturing Technology: Metal Cutting & Machine Tools, Tata Mc-